

Tic Tac Toe Game Modification Exercise

Objective:

Enhance the basic Tic Tac Toe game by adding new features and improvements. This will help you practice and solidify your skills in HTML, CSS, and JavaScript.

Instructions:

1. Change the Board Size:

- **Task:** Modify the game board from a 3x3 grid to a 4x4 grid.
- **Hints:**
 - Update the HTML to include 16 cells instead of 9.
 - Adjust the CSS grid properties (e.g., grid-template-columns) to accommodate the larger board.
 - Modify the JavaScript win-checking logic by updating the winning conditions. (For example, consider how many cells in a row you need to win on a 4x4 board.)

2. Differentiate Player Styles:

- **Task:** Style the marks for Player X and Player O differently.
- **Hints:**
 - Use CSS classes to assign different colors or animations to the cells when they are clicked.
 - Toggle these classes in the JavaScript when a cell is updated.

3. Add a Scoreboard:

- **Task:** Implement a scoreboard that tracks and displays the number of wins for each player and the number of draws.
- **Hints:**
 - Create HTML elements to display the scores above or below the game board.
 - Update these elements in your JavaScript when a game ends.
 - Optionally, include a "Reset Scores" button that clears the scoreboard.

4. Implement a Simple AI Opponent (Advanced):

- **Task:** Allow one player to play against a computer opponent.
- **Hints:**
 - After the human player makes a move, have the AI choose an available cell.

- Start with a basic implementation, such as randomly selecting one of the empty cells.
- Later, consider improving the AI to make more strategic moves.

5. **Enhance the Reset Functionality:**

- **Task:** Improve the reset button to not only clear the board but also optionally reset the scoreboard.
- **Hints:**
 - Add a confirmation prompt asking if the user wants to reset the scores.
 - Ensure that resetting the board doesn't automatically reset the scores unless confirmed.

6. **Add Visual Effects:**

- **Task:** Implement CSS transitions or animations to make the game more engaging.
- **Hints:**
 - Use CSS transitions to animate the appearance of X's and O's in the cells.
 - Add hover effects on the cells to indicate they are clickable.

Here are three highly recommended, free resources:

- **HTML:**
[MDN HTML Tutorials](#)
This resource offers a comprehensive introduction to HTML, covering the basics to advanced topics.
- **CSS:**
[MDN CSS Tutorials](#)
Explore CSS fundamentals and beyond, with clear examples and interactive demos.
- **JavaScript:**
[MDN JavaScript Tutorials](#)
Learn JavaScript step-by-step, from core concepts to more complex programming techniques.